FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2019

PAPER –1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01– 07, 19 – 25, 37 - 43) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer*
  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
  Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-C (14 – 18, 32 – 36, 50 - 54) *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

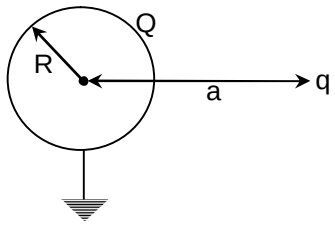
Physics

PART – I

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

1. In the given figure radius of the conducting sphere is increasing according to the relation R = vt. A point charge q is fixed at distance 'a' from centre of conducting sphere. The current flowing in the earthen conductor at the instant when R < a is



(A) $\frac{qv}{a}$ away from sphere

(B) $\frac{qv}{a}$ towards the sphere

(C) due to the reason that potential of sphere is always zero

(D) due to the reason that field inside sphere is changing with time

2. A fixed point charge Q is at origin. At t = 0, a charge q with mass m is at x = a with leftward velocity v_0 which satisfies $\frac{kQq}{a} = 3mv_0^2$. The particle turns around and starts to move rightward at the position b < a.

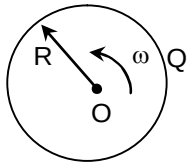
(A) the ratio b/a is 6/7

(B) the ratio b/a is 3/7

(C) the velocity of the particle at a large distance from the origin is $\sqrt{7}v_0$

(D) the velocity of the particle at a large distance from the origin is zero

3. A non – conducting disc having uniform positive charge Q, is rotating about its axis with uniform angular velocity ω. The magnetic field at the center of the disc is.



(A) directed outward

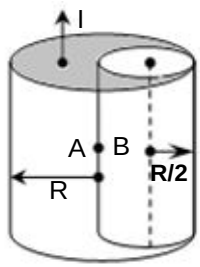
(B) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$

(C) directed inward

(D) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

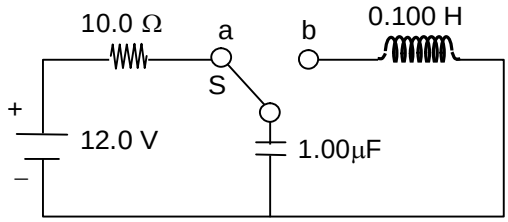
Space for Rough work

4. From a cylinder of radius R, a cylinder of radius R/2 is removed, as shown. Current flowing in the remaining cylinder is I. Magnetic field strength is



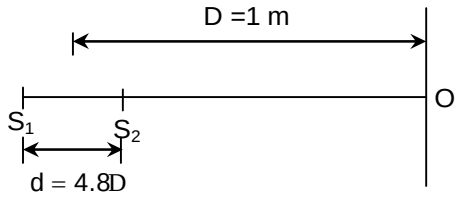
(A) zero at point A
(B) zero at point B
(C) $\frac{\mu_0 I}{3\pi R}$ at point A
(D) $\frac{\mu_0 I}{3\pi R}$ at point B

5. The switch in figure is connected to position a for a long time interval. At t = 0 the switch is thrown to position b. After this time



(A) the frequency of oscillation of the LC circuit is 200 Hz
(B) the maximum charge that appears on the capacitor is 12 μC
(C) the maximum current in the inductor is 38 mA (approx.)
(D) the total energy the circuit possesses at t = 3.00 s is 72 μJ

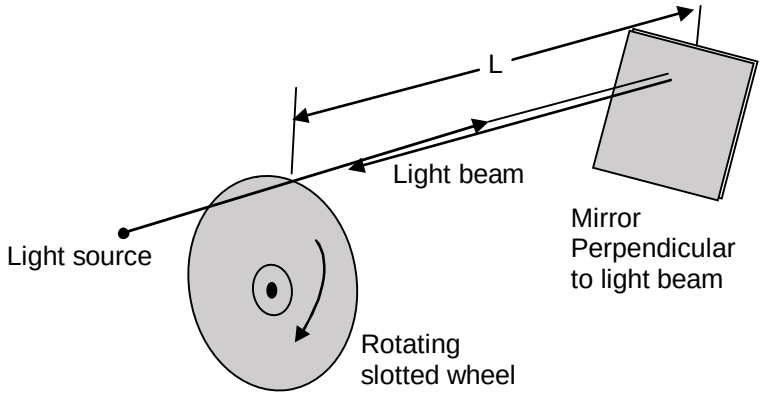
6. Two point coherent source of light of wavelength D, each placed as shown in the figure. The initial phase difference between the sources is zero (D>>d).



(A) The shape of fringes on screen is circular
(B) First minima on screen is at distance from point O is 10 cm
(C) Total minima on screen is 10
(D) Point O is a point of neither minima nor maxima

Space for Rough work

7. Figure shows an early method of measuring the speed of light that makes use of a rotating slotted wheel which rotates with constant angular velocity. A beam of light passes through one of the slots at the outside edge of the wheel, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.0 cm and 500 slots around its edge. Measurements taken when the mirror is $L = 500$ m from the wheel indicate a speed of light of 3.0×10^5 Km / s . Which of the following statements are true regarding this experiment?



(A) The angular speed of the wheel should be 1200π rad / s
(B) The angular speed of the wheel should be 1800π rad / s
(C) The linear speed of a point on the edge of the wheel should be 30π rad / s
(D) The linear speed of a point on the edge of the wheel should be 60π rad / s

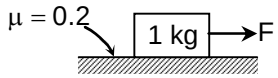
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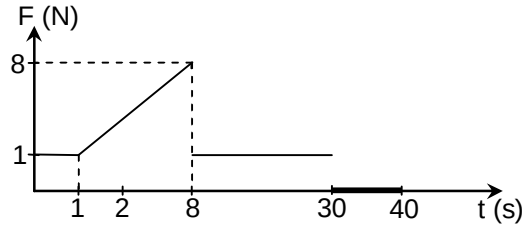
(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

A block of mass 1 kg is placed on a rough horizontal surface of coefficient of friction $\mu = 0.2$. A force is applied on the block horizontally whose variation with time is shown in the figure.





Column –I		Column –II		Column –III	
(I)	Velocity of particle	(i)	t = 1 sec	(P)	Non zero
(II)	Acceleration of particle	(ii)	t = 4 sec	(Q)	Zero
(III)	Friction force	(iii)	t = 35 sec	(R)	Greater than 1 N
(IV)	Friction force	(iv)	t = 1.5 sec	(S)	1 N

8. If we consider the velocity of the block,
(A) (I) – (i) – (Q) (B) (I) – (ii) – (Q)
(C) (I) – (iii) – (P) (D) (I) – (ii) – (P)

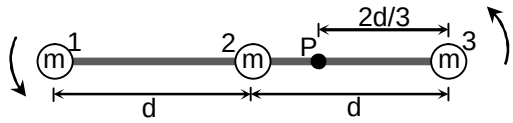
9. If we consider the acceleration of the block,
(A) (II) – (i) – (P) (B) (II) – (ii) – (P)
(C) (II) – (iv) – (P) (D) (II) – (iv) – (Q)

10. If we consider the friction force acting on the block,
(A) (III) – (ii) – (R) (B) (III) – (i) – (R)
(C) (III) – (i) – (S) (D) (III) – (iv) – (S)

Space for Rough work

Answer 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

A rigid, massless rod has three particles with equal masses attached to it as shown in figure. The rod is free to rotate in a vertical plane about a frictionless axle perpendicular to the rod through the point P and is released from rest in the horizontal position at t = 0. Assume m = 1 kg and d = 1 m.



Column –I		Column –II		Column –III	
(I)	The angular acceleration of the system (in rad/s^2)	(i)	10	(P)	Maximum
(II)	The kinetic energy of the system (in joules)	(ii)	$2\sqrt{\frac{5}{21}}$	(Q)	At t = 0
(III)	The angular momentum of the system (in kgm^2/s)	(iii)	$\frac{30}{7}$	(R)	Minimum
(IV)	The speed reached by the particle labelled 2 (in m/s)	(iv)	$2\sqrt{\frac{35}{3}}$	(S)	Zero

11. Angular acceleration of the system
(A) (I) – (ii) – (Q) (B) (I) – (i) – (Q)
(C) (I) – (iv) – (Q) (D) (I) – (iii) – (Q)

12. Kinetic energy of the system
(A) (II) – (ii) – (R) (B) (II) – (i) – (P)
(C) (II) – (iii) – (S) (D) (II) – (iv) – (P)

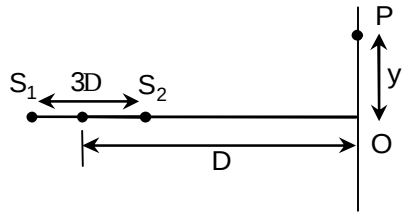
13. Angular momentum of the system
(A) (III) – (iv) – (P) (B) (III) – (ii) – (R)
(C) (III) – (i) – (R) (D) (III) – (iii) – (P)

Space for Rough work

SECTION – C
(Single digit integer type)

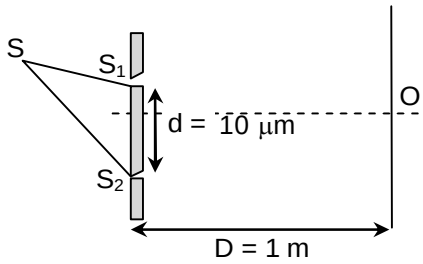
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. Figure shows two coherent microwave source S_1 and S_2 emitting waves of wavelength λ and separated by a distance 3λ. For $\lambda << D$ and $y \neq 0$, the minimum value of y for point P to be an intensity maximum is $\frac{\sqrt{m}\,\lambda D}{n}$. Determine the value of $m+n$, if m and n are coprime numbers.



15. In a Young's double slit experiment, the two slits are illuminated by a monochromatic light source S of wavelength 400 nm.

If $SS_1 = 10\ \mu m$ and SS_2 is 50 μm, find the number of maxima formed on screen.



16. A hydrogen-like atom of atomic number Z is in an excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If the makes a transition to quantum state n, a photon of energy 40.8 eV is emitted. Find the value of n.

17. Find the ratio between total acceleration of the electron in singly ionized helium atom and hydrogen atom (both in ground state).

18. Nuclei of radioactive element A are produced at rate 't^2' at any time t. The element A has decay constant λ. Let N be the number of nuclei of element A at any time t. At time $t = t_0, \frac{dN}{dt}$, is minimum. Then the number of nuclei of element A at time t = t_0 is found to be $\frac{Nt_0 - \lambda t_0^2}{\lambda^2}$. Find the value of N.

Space for Rough work

Chemistry

PART – II

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

19. Which of the following is/are ore of Cu?
(A) Chalcopyrites (B) Chalcocite
(C) Cuprite (D) Bornite

20. Which of the following is/are true statement(s)?
(A) $BCl_3 > AlCl_3 > InCl_3$ (lewis acidic strength) (B) $BF_3 < BCl_3 < BBr_3$ (Lewis acidic strength)
(C) $NH_3 < PH_3 < AsH_3$ (Reducing power) (D) $(C_2H_5)_2O > NH_3 > H_2O > PH_3$ (Bond angle)

21. Which of the following on reaction with water will produce polyprotic acid?
(A) B_2O_3 (B) N_2O_5
(C) P_4O_{10} (D) Maleic anhydride

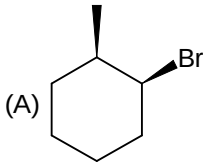
22. Which of the following reactions is/are exothermic?
(A) $SO_2 + O_2 \longrightarrow SO_3$ (B) $H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$
(C) $CO + \frac{1}{2}O_2 \longrightarrow CO_2$ (D) $B_2H_6 + O_2 \longrightarrow B_2O_3 + H_2O$

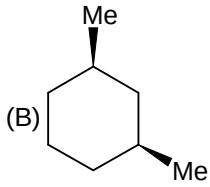
23. Xanthoproteic test is given by
(A) Tyrosine (B) Tryptophan
(C) Phenylalanine (D) Glycine

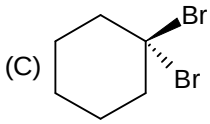
24. Lassaigne's Test for the detection of nitrogen will not fail in the case of:
(A) NH_2CONH_2 (B) $NH_2CONHNH_2$. HCl
(C) NH_2NH_2. HCl (D) $PhNHNH_2$. 2 HCl

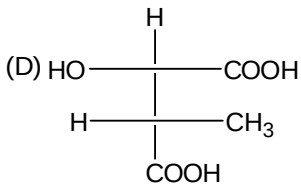
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25. Which of the following can have meso form?

(A) 

(B) 

(C) 

(D) 

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Column –I		**Column –II**		**Column –III**	
(I)	Dil. NaCl solution	(i)	After electrolysis solution becomes alkaline	(P)	One of the product is Cl_2 during electrolysis
(II)	Aq. $CuSO_4$ solution	(ii)	After electoolysis solution becomes acidic	(Q)	One of the product is H_2 during electrolysis
(III)	Pure water	(iii)	No effect on pH of solution due to electrolysis	(R)	H_2 and O_2 are the product of electrolysis
(IV)	Aq. $AgNO_3$	(iv)	Solution is acidic before electrolysis	(S)	Metal deposition at cathode and O_2 evolved at anode during electrolysis.

26. Which of the following option is correct?
(A) (IV) – (ii) – (S) (B) (IV) – (iii) – (R)
(C) (I) – (ii) – (R) (D) (II) – (i) – (S)

Space for Rough work

27. Which of the following option is correct?
(A) (II) – (iv) – (R) (B) (I) – (ii) – (R)
(C) (I) – (i) – (P) (D) (IV) – (iv) – (P)

28. Which of the following option is correct?
(A) (III) – (iii) – (R) (B) (IV) – (i) – (P)
(C) (I) – (ii) – (R) (D) (II) – (iv) – (S)

Answer 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column –I		**Column –II**		**Column –III**	
(I)	Boron $+H_2O \xrightarrow{25^\circ C} 'X'$	(i)	No reaction	(P)	Boron
(II)	$B_2H_6 + H_2O \xrightarrow{100^\circ C} y$	(ii)	Vigorous reaction	(Q)	B_2O_3 is the product
(III)	$I_4O_9 + H_2O \longrightarrow P + Q$	(iii)	Acid/s are the product	(R)	'P' or 'Q' + ethylene glycol $\rightarrow$ formaldehyde
(IV)	$Cl_2O_7 + H_2O \longrightarrow R$	(iv)	Base/s are the product	(S)	R + Benzaldehyde $\rightarrow$ benzoic acid

29. Which of the following option is correct?
(A) (I) – (ii) – (P) (B) (II) – (ii) – (Q)
(C) (III) – (iii) – (R) (D) (IV) – (iii) – (S)

30. Which of the following option is correct?
(A) (I) – (i) – (P) (B) (II) – (ii) – (R)
(C) (III) – (iii) – (P) (D) (III) – (iv) – (Q)

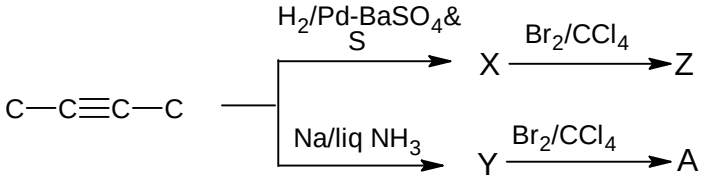
31. Which of the following option is correct?
(A) (III) – (iii) – (R) (B) (II) – (iv) – (P)
(C) (I) – (i) – (P) (D) (I) – (ii) – (Q)

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. Calculate Z + A.



33. For the reaction $2NO + H_2 \rightarrow N_2O + H_2O$ at 900 K, following data are observed.

Initial Pressure of NO	Initial pressure of H_2	Rate
0.15	0.40	0.02
0.075	0.40	0.005
0.150	0.02	0.10

The order of reaction is

34. The pyrolysis of an ester follows a first order process and its rate constant can be expressed as $In\,k = 78.09 - \frac{42075}{T}$. where k is given in the min^{-1}. The time required for 25% reaction to complete at 227^0 C will be _____ hr.

35. From the given compounds number of compounds that will give positive carbylamine test will be____.
N, N dimethyl Aniline
2, 4 Dimethyl Aniline
N- methyl – o – methyl Aniline
O – methyl benzyl Aniline

36. Number of ambident nucleophlie from the following are_____
—ONO , —OCH_3 , —CN, —CNO.

Space for Rough work

Mathematics

PART – III

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

37. $\lim_{n\to\infty}\sum_{K=0}^{K=n}\frac{{}^{n}C_{k}}{n^{K}(K+3)} = a+be$ Where $a,b \in I$ and "e". Natural base lies between 2 & 3. Then, correct option is

(A) $|a|+|b| = a^{|a|}+1$ (B) $|a+b| = 1$

(C) $|a|+|b| = 3$ (D) $|a-b| = |b|^{a}+|a|$

38. If $f(x) = x^2e^{-x} + \int_0^x e^{-t}f(x-t)dt$ and $g(x) = \left(\frac{2-f(x)}{2}\right)$ then

(A) $\frac{2-f(\ln 5)}{1+\ln 5} = 2/5$ (B) $\frac{2-f(\ln 5)}{1+\ln 5} = 5/2$

(C) $g(x) \le 1$ (D) $g(x) \ge -1$

39. If $f(x) = \frac{1+x^2}{1+x}$ for $x \ge 0$ the least value of f(x) is β at $x = \alpha$, then

(A) $\beta = 2\alpha$ (B) $\beta = 2\alpha + 1$

(C) $\alpha^2 + \beta = 1$ (D) $|\alpha^2 - \beta| = 1$

40. If $5x\left(1+\frac{1}{x^2+y^2}\right) = 12$ and $5y\left(1-\frac{1}{x^2+y^2}\right) = 4$ $\forall x,y \in R$, then correct option is/are

(A) $|x|+|y| = 3$ (B) $x+y = \frac{1}{5}$

(C) $|x|+|y| = \frac{3}{5}$ (D) $x^2+y^2 = 5$

Space for Rough work

41. If $f(x)=\dfrac{\cos^{-1}\left(1-\{x\}\right).\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}.\left(1-\{x\}\right)}$ here {.} denotes fractional part of x then;

(A) $\underset{x\to 0^+}{\text{Lt}}\, f(x)=\sqrt{2}\underset{x\to 0^-}{\text{Lt}}\, f(x)$ (B) $\underset{x\to 0^-}{\text{Lt}}\, f(x)=\sqrt{2}\underset{x\to 0^+}{\text{Lt}}\, f(x)$

(C) $\underset{x\to 0^-}{\text{Lt}}\, f(x)=\dfrac{\pi}{2\sqrt{2}}$ (D) $\underset{x\to 0^-}{\text{Lt}}\, f(x)=\pi\sqrt{2}$

42. If $\int_a^b|\sin x|dx=8$ & $\int_0^{a+b}|\cos x|dx=9$ then;

(A) $a+b=9\pi$ (B) $b-a=4\pi$

(C) $\int_a^b x\sin x dx=2\sqrt{2}\pi$ (D) $\int_a^b x\sin x dx=-2\sqrt{2}\pi$

43. If $\Delta(x)=\begin{vmatrix}\cos x+1 & \cos^2 x & 1+\sin^2 x\\ -\sin x & -\sin 2x & \sin 2x\\ \sin x & 2^2\sin 2x & -2^2\cos 2x\end{vmatrix}$, then

(A) $\int_0^{\pi/2}\Delta(x)dx=4$ (B) $\int_0^{\pi/2}\Delta(x)dx=-4$

(C) $\int_{\pi/2}^{\pi}\Delta(x)dx=4$ (D) $\int_{\pi/2}^{\pi}\Delta(x)dx=-4$

Space for Rough work

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Consider four distinct positive real numbers a, b, c and d.

Column –I		**Column –II**		**Column –III**	
(I)	a, b, c, d are in A.P.	(i)	$ab + cd > 2bc$	(P)	$ac < bd$
(II)	a, b, c, d are in G.P.	(ii)	$bc > ad$	(Q)	$ab + cd > 2ad$
(III)	ab, bc, cd are in G.P.	(iii)	$ab + bc < ad$	(R)	$ab + bc + cd > 3ad$
(IV)	ab, bc, cd are in H.P.	(iv)	$a + d > b + c$	(S)	$a + c < b + d$

44. Which of the following options is correct?
(A) (I) – (iii) – (Q) (B) (I) – (ii) – (R)
(C) (IV) – (iii) – (P) (D) (III) – (ii) – (R)

45. Which of the following options is correct?
(A) (II) – (iv) – (R) (B) (I) – (iii) – (P)
(C) (III) – (ii) – (R) (D) (IV) – (iii) – (Q)

46. Which of the following options is correct?
(A) (I) – (ii) – (S) (B) (II) – (iv) – (P)
(C) (III) – (i) – (Q) (D) (III) – (ii) – (R)

Space for Rough work

Answer 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

Column –I		**Column –II**		**Column –III**	
(I)	If $\lim_{x\to\infty}\left(\frac{x^2+1}{x+1}-ax-b\right)=0$, then	(i)	a = 2	(P)	$b \in R$
(II)	$\lim_{x\to 0}\left(1+ax+bx^2\right)^{\frac{2}{x}}=e^3$, then	(ii)	$a=\frac{3}{2}$	(Q)	b = 2
(III)	$\lim_{x\to 0}\left(\frac{ae^x-b}{x}\right)=2$	(iii)	a = 1	(R)	b = –1
(IV)	$\lim_{x\to-\infty}\left\{\sqrt{\left(x^2-x+1\right)}-ax-b\right\}=0$	(iv)	a = –1	(S)	$b=\frac{1}{2}$

47. Which of the following is correct?
(A) (I) – (ii) – (R) (B) (I) – (iii) – (R)
(C) (I) – (iii) – (S) (D) (I) – (iv) – (P)

48. Which of the following is correct?
(A) (II) – (ii) – (P) (B) (II) – (ii) – (R)
(C) (II) – (iii) – (Q) (D) (II) – (iv) – (S)

49. Which of the following is correct?
(A) (III) – (ii) – (R) (B) (III) – (iv) – (R)
(C) (III) – (i) – (Q) (D) (III) – (ii) – (S)

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If $x+y=3$ is tangent at (2, 1) on hyperbola, intersects asymptotes at A and B such that $AB=8\sqrt{2}$. If centre of hyperbola is (–1, –1), then least possible value of sum of semi-transverse axis and semi-conjugate axis is $p\sqrt{q}$, where HCF (p, q) = 1 and $\sqrt{q}$ is irrational number. Then, $(p+q)$ is equal to

51. If θ is acute angle of intersect between two curves $x^2+5y^2=6$ and $y^2=x$, then $\cot\theta$ is equal to

52. If 8 harmonic means are inserted between two numbers a and b (a < b) such that arithmetic mean of a and b is $\frac{5}{4}$ times equal to geometric mean of a and b, then $\left(\frac{H_8-a}{b-H_8}\right)$ is equal to

53. If y = 2 is directrix and (0, 1) be the vertex of parabola $x^2+\lambda y+\mu=0$ then; $(\lambda-\mu)=?$

54. The minimum number of real roots of $\left(x^2+3x+a\right)\left(x^2+ax+1\right)=0$ equals

Space for Rough work